

Standard Operating Procedure

Shimadzu TOC Analyzer with SSM for
Soil Samples with Varying Carbonate Concentrations

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1 Purpose and Scope

This SOP describes the procedure for determining Total Carbon (TC), Total Inorganic Carbon (TIC), and Total Organic Carbon (TOC) in soil samples using the Shimadzu TOC Analyzer equipped with a Solid Sample Module (SSM). This procedure is applicable to soil samples with high ($> 10\%$), medium ($2\text{--}10\%$), and low ($< 2\%$) carbonate concentrations.

2 Principle

The Shimadzu TOC-SSM system uses high-temperature combustion (900°C) to convert all carbon in the sample to CO_2 , which is then quantified by NDIR (Non-Dispersive Infrared) detection. For carbonate-bearing soils:

- **TC (Total Carbon)** = measured by direct combustion
- **TIC (Total Inorganic Carbon)** = measured by acid treatment (200°C)
- **TOC (Total Organic Carbon)** = $\text{TC} - \text{TIC}$

3 Safety and Precautions

3.1 Personal Protective Equipment (PPE)

- Safety glasses or goggles
- Laboratory coat
- Heat-resistant gloves when handling hot ceramic boats
- Nitrile gloves when handling acids

3.2 Chemical Hazards

- Phosphoric acid (25% solution): corrosive, causes severe burns
- High-purity oxygen: oxidizer, keep away from combustibles
- Soil samples: may contain pathogens or heavy metals

3.3 Equipment Hazards

- Combustion furnace reaches 900°C
- Acid furnace reaches 200°C
- Allow adequate cooling time before handling boats

4 Equipment and Materials

4.1 Instrumentation

- Shimadzu TOC Analyzer with Solid Sample Module (SSM)
- High-purity oxygen supply (99.999%)
- Carrier gas supply (oxygen or nitrogen, 99.999%)
- Computer with TOC Control Software
- Analytical balance (0.0001 g precision)

4.2 Consumables

- Ceramic sample boats (pre-combusted)
- 25% phosphoric acid solution (prepared fresh weekly)

- Glucose or sucrose standard (certified reference material)
- Calcium carbonate standard (reagent grade, dried at 105°C)

4.3 Sample Preparation Equipment

- Mortar and pestle or ball mill
- 2 mm and 0.25 mm sieves
- Drying oven (105°C)
- Desiccator with desiccant

5 Reagents and Standards

5.1 Phosphoric Acid Solution (25% v/v)

1. Add 250 mL concentrated phosphoric acid (85%) to 600 mL deionized water
2. Mix thoroughly and dilute to 1000 mL
3. Store in glass bottle, prepare fresh weekly

5.2 TOC Calibration Standards

Prepare from glucose or sucrose:

- Stock solution: 1000 mg C/L
- Working standards: 10, 25, 50, 100, 250, 500 mg C/L
- Prepare fresh daily

5.3 TIC Calibration Standards

Prepare from dried calcium carbonate (CaCO_3):

- Calculate carbon content: 12.0% C by weight
- Prepare solid standards: 1, 5, 10, 25, 50 mg CaCO_3
- Store in desiccator

6 Sample Preparation

6.1 Initial Processing

1. Air-dry soil samples at room temperature or 40°C maximum
2. Remove large debris, roots, and stones
3. Grind samples using mortar and pestle or ball mill
4. Pass through 2 mm sieve for initial homogenization

6.2 Fine Grinding

1. Take representative subsample (approximately 50 g)
2. Further grind to pass through 0.25 mm sieve (60 mesh)
3. Mix thoroughly to ensure homogeneity

6.3 Moisture Determination

1. Weigh 2–5 g of ground sample in pre-weighed crucible
2. Dry at 105°C for 24 hours
3. Cool in desiccator and reweigh
4. Calculate moisture content for dry weight correction

6.4 Sample Weighing Guidelines

Soil Type		TC Analysis	TIC Analysis	Expected TIC
High	Carbonate	10–20 mg	5–10 mg	> 1.2% C
(> 10%)				
Medium	Carbonate (2–	20–40 mg	10–30 mg	0.24–1.2% C
10%)				
Low	Carbonate (< 2%)	40–80 mg	30–80 mg	< 0.24% C

7 Instrument Setup and Calibration

7.1 Daily Startup

1. Turn on carrier gas and oxygen supply
2. Check gas pressures (typically 0.4–0.5 MPa)
3. Power on TOC analyzer and allow 30-minute warm-up
4. Launch TOC Control Software
5. Set furnace temperatures:
 - TC furnace: 900°C
 - TIC furnace: 200°C
6. Verify baseline stability (< 5 ppm drift)

7.2 System Blank Check

1. Analyze 3 blank ceramic boats
2. Average blank should be < 0.1 mg C
3. If blank > 0.1 mg C, clean boats or replace

7.3 TC Calibration

1. Analyze glucose/sucrose standards in triplicate
2. Use minimum 5 concentration levels
3. Generate calibration curve ($r^2 > 0.999$ required)
4. Verify linearity across working range

7.4 TIC Calibration

1. Weigh CaCO_3 standards directly into ceramic boats
2. Add 150 μL of 25% phosphoric acid
3. Analyze in triplicate at each concentration
4. Generate calibration curve ($r^2 > 0.998$ required)

7.5 Quality Control Standards

1. Analyze certified reference soil (if available)
2. Expected recovery: 90–110%
3. If outside range, recalibrate system

8 Analysis Procedure

8.1 Sample Loading

1. Pre-clean ceramic boats by combusting at 900°C
2. Cool boats in desiccator for minimum 30 minutes
3. Tare balance with empty boat
4. Add appropriate sample amount according to carbonate level
5. Record exact sample weight (to 0.0001 g)
6. Place boat in SSM autosampler carousel

8.2 TC Analysis

1. Create sample sequence in software
2. Enter sample IDs and weights
3. Select TC analysis method:
 - Combustion temperature: 900°C
 - Combustion time: 3–5 minutes
 - Oxygen flow: per manufacturer specifications
4. Start analysis sequence
5. System automatically combusts each sample
6. Software calculates TC based on calibration curve

8.3 TIC Analysis

1. Prepare new sample boats with same soil samples
2. Weigh samples (use same amounts as TC analysis)
3. Add 150 μL of 25% phosphoric acid to each boat
4. Wait 30 seconds for initial reaction
5. Load boats into TIC furnace position
6. Select TIC analysis method:
 - Temperature: 200°C
 - Analysis time: 5–7 minutes
 - Oxygen/carrier gas flow: per manufacturer specifications
7. Start analysis sequence
8. Acid converts carbonates to CO_2 at low temperature
9. Software calculates TIC based on calibration curve

8.4 TOC Calculation

$$TOC(\%) = TC(\%) - TIC(\%) \quad (1)$$

Software typically calculates automatically.

9 Quality Assurance/Quality Control

9.1 Replicate Analysis

- Analyze each sample in triplicate minimum
- Calculate mean, standard deviation, and RSD
- Acceptable RSD: < 10% for samples > 0.5% C
- Acceptable RSD: < 15% for samples < 0.5% C

9.2 Calibration Verification

- Analyze calibration check standard every 10 samples
- Recovery must be 90–110%
- If outside range, recalibrate and re-analyze affected samples

9.3 Method Blanks

- Analyze blank boat every 10 samples
- Blank should be < 10% of lowest sample concentration

9.4 Duplicate Samples

- Analyze 10% of samples as duplicates
- RPD (Relative Percent Difference) should be < 15%

9.5 Certified Reference Materials

- Analyze CRM at beginning and end of each run
- Document recovery and control chart results

10 Carbonate-Specific Considerations

10.1 High Carbonate Soils (> 10%)

10.1.1 Challenges:

- Vigorous reaction with acid
- Potential for sample loss or splashing
- May exceed detector range

10.1.2 Solutions:

- Use smaller sample sizes (5–10 mg for TIC)
- Add acid slowly in 50 μL increments
- Wait 1–2 minutes between acid additions
- Dilute samples if necessary
- Consider pre-treatment to remove excess carbonates for TOC determination

10.2 Medium Carbonate Soils (2–10%)

10.2.1 Considerations:

- Standard protocol usually sufficient
- Monitor for complete carbonate dissolution
- Ensure adequate acid addition (150–200 μL)
- Extend TIC analysis time if needed (7–10 minutes)

10.3 Low Carbonate Soils (< 2%)

10.3.1 Considerations:

- Use larger sample sizes to improve detection
- TIC may be near detection limit
- Verify blank contamination is minimal

- Multiple replicates recommended ($n = 5$)
- TOC may be determined by direct method after acid pre-treatment

10.4 Acid Pre-treatment for Direct TOC (Optional)

For high carbonate soils, direct TOC determination:

1. Weigh 0.5–1 g soil into beaker
2. Add 10 mL of 1M HCl dropwise
3. Allow to react until effervescence stops
4. Repeat acid addition if necessary
5. Rinse with deionized water ($3\times$)
6. Dry at 60°C
7. Grind and analyze for TOC directly
8. **Note:** This method may result in loss of volatile organic compounds

11 Data Recording and Calculations

11.1 Raw Data

Record in laboratory notebook:

- Sample ID and description
- Sample weight (dry weight basis)
- Analysis date and time
- Instrument ID and calibration date
- Peak area/height
- Calculated concentrations

11.2 Calculations

11.2.1 Dry weight correction:

$$\text{Dryweight} = \text{Wetweight} \times \left(1 - \frac{\text{Moisture}\%}{100}\right) \quad (2)$$

11.2.2 Carbon concentration (%):

$$C(\%) = \frac{C_{\text{measured}}(\text{mg})}{S_{\text{sample dry weight}}(\text{mg})} \times 100 \quad (3)$$

11.2.3 TOC calculation:

$$\text{TOC}(\%) = \text{TC}(\%) - \text{TIC}(\%) \quad (4)$$

11.2.4 Relative Percent Difference (RPD):

$$\text{RPD} = \frac{|Sample1 - Sample2|}{\left(\frac{Sample1 + Sample2}{2}\right)} \times 100 \quad (5)$$

11.3 Reporting

Report results to appropriate significant figures:

- TC, TIC, TOC: 2 decimal places
- Include standard deviation or RSD
- Report on dry weight basis
- Note any dilutions or pre-treatments

12 Maintenance and Troubleshooting

12.1 Daily Maintenance

- Check gas pressures and flows
- Verify furnace temperatures
- Clean sample inlet if needed
- Check for leaks in gas lines

12.2 Weekly Maintenance

- Clean combustion tubes
- Replace quartz wool if discolored
- Check and clean NDIR detector cell
- Verify calibration standards

12.3 Monthly Maintenance

- Replace oxygen scrubber cartridge
- Check all O-rings and seals
- Clean or replace furnace tubes if necessary
- Perform full system calibration

12.4 Common Issues and Solutions

Problem	Possible Cause	Solution
High blank values	Contaminated boats	Pre-combust boats at 900°C
Poor peak shape	Incomplete combustion	Increase combustion time/temperature
Low TIC recovery	Insufficient acid	Increase acid volume or add incrementally
Carryover between samples	Incomplete purging	Extend purge time between samples
Baseline drift	Detector contamination	Clean NDIR cell
Inconsistent replicates	Sample heterogeneity	Improve grinding and mixing

13 Waste Disposal

- Acid waste: Neutralize with sodium bicarbonate, dispose per institutional guidelines
- Ceramic boats: Reuse after cleaning or dispose as non-hazardous solid waste
- Soil samples: Dispose per institutional chemical waste procedures

14 References

1. Shimadzu TOC Analyzer SSM-5000A User Manual
2. ASTM D4129 - Standard Test Method for Total and Organic Carbon in Water
3. ISO 10694 - Soil Quality - Determination of Organic and Total Carbon after Dry Combustion
4. Nelson, D.W. and Sommers, L.E. (1996) Total Carbon, Organic Carbon, and Organic Matter. Methods of Soil Analysis Part 3

15 Revision History

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1.0	October 2025	Initial release	

16 Approval

Role	Name	Signature	Date
Author			
Reviewer			
Approver			